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MATH

for Data Science

What do you need to know?

Swipe to find out

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LINEAR ALGEBRA

- Scalars, vectors, matrices, and tensors
- Determinant, transpose, and inverse of a matrix
- Linear transformations
- Dimensionality reduction
  - Principal Component Analysis (PCA)
  - Linear Discriminant Analysis (LDA)

CALCULUS

- Differentiation
  - Gradient descend
  - Back propagation

Integration

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STATISTICS

Descriptive statistics

- Numerical and categorical variables
- Mean, median, and mode
- Skewness
- Variance and covariance
- Correlation
- Standard deviation

Inferential statistics

- Normal and standard normal distribution
- Central limit theorem
- Standard error

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STATISTICS

Hypothesis testing

- Null and alternative hypothesis
- Type I and Type II errors
- p-values

Statistical models and methods

- Simple linear regression
- Least squares and maximum likelihood
- Logistic regression

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PROBABILITY

- **Combinatorics**
  - Permutations
  - Variations
  - Combinations
- **Sets and events**
  - Intersection and union of sets
  - Mutually exclusive sets
  - Dependent and independent events
  - Conditional probability
- **Bayes rule**

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PROBABILITY

Discrete distributions

- Uniform distribution
- Bernoulli distribution
- Binomial distribution
- Poisson distribution

Continuous distributions

- Normal distribution
- Students T distribution
- Chi-squared distribution
- Exponential distribution
- Logistic distribution